The Wesleyan Science Journal
Journal Content & Editorial Calendar

The Wesleyan Science Journal will serve as a hybrid model between a traditional peer-reviewed scientific journal and a science magazine. As such, we aim to publish a variety of content, including journal articles, review articles, interviews, and personal essays. The layout and content separation are subject to change as the *Wesleyan Science Journal* evolves. Our content will be sorted into two overarching themes: **Research & Reviews** and **News, Features, & Perspectives**

Research & Reviews:

Journal Articles:

1. Purpose: Journal articles are peer-reviewed* scientific papers detailing original research.
2. Structure:
    a. Abstract (1-4 sentences): This is a snapshot of your research study, with each sentence describing the major takeaway from each of the following sections.
    b. Introduction (1-2 paragraphs): Introduce your topic and the relevant information needed to understand both your study and research question.
    c. Methods/Materials (1 paragraph): Describe your experimental setup and procedure.
    d. Results (1-2 paragraphs): Share your results (graphics are particularly helpful here), but be sure not to begin any analysis.
    e. Discussion (3-4 paragraphs): Analyze your results and contextualize your findings in the larger scope of the field you are studying. Be sure to highlight limitations of your study and future avenues for research.
    f. Acknowledgements: This is where you can thank your research mentors and any peers that helped you out throughout the research process.
    g. References: A list of all sources used in your research and writing process. Citation style will vary between disciplines, so follow the standard citation style for your given field.
3. Additional Guidance: If you are unsure about the structure, tone, or style of your writing, be sure to read published papers in your given discipline as a guide to build your own style of scientific writing. Be sure to always cite your sources, never plagiarize, and maintain scientific integrity as you write.

*While the *Wesleyan Science Journal* will not utilize a *formal* peer review process, all original research will need to be vetted by the appropriate editorial staff, as well as the Editors-in-Chief. Prior to publication, the corresponding Principal Investigator (PI) will need to provide permission for the display and distribution of all original research.

News and Views:

1. Purpose: News and Views articles are a more accessible mode of accessing original scientific research, as they serve to summarize the findings of a particular paper in the larger context of their particular field of study.
2. **Structure:
    a. Background (1-2 paragraphs): Describe the topic/field of interest of this particular paper, taking care to make sure the language and content are broadly accessible to other researchers in that discipline who do not research the same topic.
    b. Paper Introduction (1-2 paragraphs): Now is your time to introduce the specifics of your paper and the specific research question that is being investigated.
    c. Important Findings (4-8 paragraphs): Get into the specifics of this paper, highlighting experiments and experimental results that are particularly relevant and compelling. When determining what information to highlight, consider this: What makes this paper particularly important to this topic? What differentiates it from past research that has been done?
    d. Larger Relevance (1-2 paragraphs): Now get into the potential impact the findings from this study will have both on this specific research, but also on that field of study at large.
    e. Further Studies: This section is largely optional, but if papers have been published following your paper that have built on/been influenced by this research, feel free to spend some time highlighting them.
    f. References: A list of all sources used in your research and writing process. Citation style will vary between disciplines, so follow standard citation style for your given field.
3. Additional Guidance: Journals such as *Nature* and *Science* also contain News and Views (or similarly titled sections) and are a great resource to turn to when determining stylistic choices and structuring content.

** These guidelines are adapted from an assignment description from Professor Jennifer Mitchel's BIOL 267.

Literature Review Articles:

1. Purpose: Literature reviews summarize and evaluate published research on a given topic and serve to identify future avenues of research within a particular field.
2. Structure:
    a. Background and Introduction (1–2 paragraphs): Introduce the broader topic, explain why it is important, and provide enough background for readers in the same general discipline who may not specialize in this area. Clearly identify the focus of the review (question, debate, problem, development, etc.).
    b. Main Discussion (5–10 paragraphs): Organize the research into major themes, methods, theories, or debates. Compare findings across studies and explain how they relate to one another. Avoid summarizing each paper separately; instead, use multiple sources to support each main idea. There are different organizational approaches such as: thematically, chronologically, methodologically, theoretically, etc. Choose the approach that best fits your topic and research paper.

c. Gaps and Future Directions (1–3 paragraphs): Identify important unanswered questions, limitations, or inconsistencies in the literature. Explain how future research could address them.
d. Conclusion (1–2 paragraphs): Summarize what the literature collectively suggests and explain the broader significance of the topic.
e. References: A list of all sources used in your research and writing process. Citation style will vary between disciplines, so follow standard citation style for your given field.

3. Additional Guidelines: Here is some helpful advice on writing a literature review along with other sources online:

- How to write a superb literature review
- UNC Writing Center — Literature Reviews
- Purdue OWL — Writing a Literature Review
- Nature Reviews — Author Guidelines
- Science — Instructions for Preparing an Initial Manuscript

News, Features & Perspectives

News:

1. Purpose: News articles serve as an opportunity to tell the story of a given research lab, study, event, or other noteworthy scientific news within the Wesleyan community. **NOTE:** This category is **distinct** from News & Views in the *Research & Reviews* section. News articles report on timely events and developments, whereas News & Views articles analyze and contextualize published scientific research.
2. Structure:
    a. Lead: The lead is the opening paragraph of the article, and it contains the crucial background information: who, what, when, where, why.
        i. Remember, your lead is the first part of your article that your reader is going to see, so make sure it draws them into the story:
        ii. There are several ways to write a compelling lead. Here's a good resource to look through: A good lead is everything — here's how to write one
    a. Body Paragraphs: This is the bulk of your article, and where you effectively tie together the quotes from your interviews to piece together a larger narrative.
    b. Tail/Conclusion: The tail helps finish your article by providing an interesting quote, final thought, etc. to leave your reader with.
    c. Additional Guidelines:

Features:

1. Purpose: Feature articles provide an opportunity to explore scientific topics, people, and experiences in a more in-depth and narrative way than traditional news articles.
    a. Unlike news stories, features are not necessarily tied to recent events and may focus on the broader context, human stories, or significance of a topic.

2. Structure: There is no single structure for feature articles. Depending on the subject, a feature may take the form of a profile, research spotlight, lab spotlight, behind-the-scenes story, explainer, or another creative format.
    a. Writers are encouraged to organize their article in whatever way best tells the story while maintaining clarity and accessibility.
    b. Feature articles may range from 500 to 2,000 words, depending on the subject matter and writing style. If you are considering a piece outside of this range, please discuss it with staff.
    c. Additional Guidelines: Feature articles are intended for a broad scientific audience and should balance engaging storytelling with scientific accuracy. Unlike other articles, there is no prescribed structure for features; writers are encouraged to choose the format that best tells their story while remaining clear and accessible. The term *feature* is intentionally broad, and we welcome creative approaches to science communication. If you have an idea for an article that does not fit neatly into another category, discuss it with your editorial staff—we're always open to new ideas.
        i. Here are some ideas to get you started:
            1. Research Lab features
            2. Historical features
            3. Science culture (What happens before a paper is published? What happens at scientific conferences?)

Op/Ed:

1. Purpose: Opinion pieces are effectively argumentative essays, and for our journal can serve to highlight important debates/discourse that is currently going on in the Wesleyan scientific community, and thereby the scientific community at large
2. Format:
    a. Lead: Same as News, the lead is the opening paragraph of the article, and it contains the crucial background information: who, what, when, where, why.
    b. Thesis: Your thesis is your overarching argument for your paper. Be sure to make it clear and concise.
    c. Argument 1: Your first piece of evidence supporting your thesis
    d. Argument 2: Your second piece of evidence supporting your thesis
    e. Argument 3: Your third piece of evidence supporting your thesis
    f. Counter-argument: Every strong opinion piece acknowledges the opposing side. Use the counterargument as a way to recognize potential criticisms, and use it to ultimately fortify the strength of your overall argument.
    g. Conclusion: Hit it home with a compelling call to action that ties into your original thesis.
3. Additional Guidelines: Our Op/Ed guide is adapted from The Op/Ed Project, but feel free to play around with the structure to fit the individual needs of your argument. Also, turn to the Opinion section of publications such as *Nature* and *Science* to explore writing styles and structure formats.

Editorial Calendar

Article Meetings & Brainstorming:

- **Article Meetings:** serve as an opportunity for editors to converse with one another to set the article list for their section.
    - Lead Editors will start the brainstorming process by opening a dialogue between editors that will continue into the actual article meeting, by the end of which the article list will be set (pending approval from the Lead Editors and Editors-in-Chief).
- **Brainstorming**: prior to meeting with your editorial team, spend some time thinking and talking about events that are at the forefront of your scientific discipline.
    - Research & Reviews:
        - Find labs, faculty members, and student researchers within your department that you want to learn more about! Let your own interests guide your search for an article
    - News, Features & Perspectives
        - Talk to your professors and peers! They are an excellent resource and can help you start finding events, people, or even student groups to write about.
        - Stay up to date with the departmental newsletters to keep track of events
        - Find Wes alumni within your field that you think would make for interesting interviews, especially to highlight lesser-known career paths or especially interesting opportunities.

Interviews:

Interviews are the backbone of news! A good interview can make or break your article, so here are some tried-and-true tips:

- **Contact interviewees early.** If you want to speak with a faculty member, student researcher, or alumnus, you must give yourself ample time to write interview questions, reach out, receive a response, schedule a time to meet, and write your article.
- **Conduct interviews in a professional manner.** For all in-person (or Zoom) interviews, arrive early with pre-prepared questions and suitable recording devices.
    - Make sure to get permission from your interviewee before recording.
    - In-person interviews are strongly encouraged, but in circumstances where meeting is not possible, you can send pre-written questions over email. If this is the case, contact your Lead Editor and the Editors-in-Chief to let them know.
    - Remember, interviews are a great opportunity to connect with and learn from experienced individuals within the Wesleyan community who share your interests and disciplines, so be engaged!
- **Let interviews be a conversation.** While you always should be prepared with questions for your interviewees, try not to come in with any preconceived notions about your piece. Use questions to guide the conversation, not limit it.

Article Edits:

- **Editing:** As we will generally have articles due one week before the layout meeting, we expect all members of the editorial team to efficiently and thoughtfully read, edit, and provide feedback on the articles within their section
    - Don't just look for grammatical errors; think critically about the piece as a whole and try to read the article from the perspective of someone who isn't as knowledgeable on the subject. Remember, our hope as a journal is to provide rigorous scientific research and news that can be accessed by science and non-science readers on campus alike.
- **Timeliness**: Once you submit your articles, you will spend the next 3-4 days not only making edits on other people's writing but accepting edits for your own. The earlier you all do your edits, the earlier you can adjust your piece as necessary.

Layout:

- **Layout:** We will all meet as an editorial team to lay out articles using InDesign.
    - Print Journal Structure: As mentioned above, the journal layout will be split into two overarching categories: **Research & Reviews** and **News, Features & Perspectives**
        - **News, Features & Perspectives** (15-20 pages)
            - Life Science
            - Physical Science
            - Quantitative & Computational Science
            - Science, Technology & Society
        - **Research & Reviews** (15-20 pages)
            - Life Science
            - Physical Science
            - Quantitative & Computational Science
            - Science, Technology & Society

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